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Bioavailability and methylation of bulk mercury sulfide in paddy soils: New insights into mercury risks in rice paddies

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ABSTRACT

Mercury sulfide (HgS) constitutes the largest Hg reservoir in the lithosphere but has long been considered to be not bioavailable and a minor participant in Hg biogeochemical cycling. Herein, we report that bulk α-HgS can be dissolved and methylated in paddy soils, especially with rice culture. Bulk α-HgS exposure did not affect rice growth compared to the control group but significantly increased methylmercury (MeHg) contents in the rhizospheric soils (e.g., 80.15% and 108.71% higher for bulk α-HgS treatment vs. control at the seedling and maturation stages, respectively). Moreover, bulk α-HgS exposure induced an apparent accumulation of MeHg (50% higher for bulk α-HgS treatment vs. control) and markedly elevated total Hg (THg) in rice grains. The presence of DOM and reduced sulfide in paddy soils was further evidenced to drive the mobilization and dissolution of bulk α-HgS, thereby resulting in THg and MeHg accumulation in rice grains. These findings highlight the bioavailability of HgS in rice paddies and suggest that bulk HgS should be considered when assessing Hg health risks and developing efficient remediation approaches in Hg-contaminated croplands.

1. Introduction

The toxicity of mercury (Hg) is closely associated with its speciation. Methylmercury (MeHg) is the most toxic Hg species that can cross the blood-brain barrier and be enriched in the human body, resulting in health risks (Díez, 2009; Syversen and Kaur, 2012; Beckers, Rinklebe, 2017). Dietary MeHg exposure leads to the mental impairment of 200,000–1800,000 newborns in EU countries each year and a decrease of 0.145 in the average IQ of newborns in China (Chen et al., 2019). MeHg intake can also cause male infertility (Wang et al., 2010). The consumption of contaminated marine products is generally considered the main pathway of MeHg exposure for humans. However, recent studies indicated a large production of MeHg in paddy fields and a high accumulation of MeHg in rice grains due to the enhanced growth of anaerobic methylated microorganisms, such as sulfate-reducing bacteria, iron-reducing bacteria, and methanogens, in Hg-contaminated paddy fields (Zhao et al., 2016; Gilmour et al., 2013; Zhu et al., 2017; Liu et al., 2018; Rothenberg and Feng, 2012). In a previous investigation, Hg contents in 560 rice samples collected from 15 provinces across China showed lower values than the national Hg limited standard (20 μg/kg), which indicated low exposure risks from rice consumption for the general population in these investigated regions (Zhao et al., 2019). However, it has been reported that the concentrations of total Hg (THg) and MeHg in rice grains collected in Wanshan Hg mining areas (Guizhou, China) are up to 569 μg/kg and 145 μg/kg, respectively, which are much higher than the national standard for cereals (20 μg/kg, GB2762–2017) (Horvat et al., 2003; Qiu et al., 2008). Moreover, more than 12% of the rice samples (among 303 rice samples collected from 9 provinces in China) contained Hg exceeding the national Hg limited standard (20 μg/kg) (Yuan et al., 2011). These works suggest the uneven distribution and typically regional contamination of Hg in China. In Hg-contaminated areas, rice consumption is believed to be an important source of THg/MeHg intake, specifically contributing 7 – 96% of dietary MeHg exposure, and increasing the health risks for local residents (Meng

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et al., 2014; Feng et al., 2008; Gong et al., 2018; Zhang et al., 2010).

The production of MeHg in paddy fields is closely associated with inorganic Hg speciation and the activity of Hg-methylating microorganisms. The chemical forms of soil inorganic Hg include soluble/dissoluble and exchangeable Hg, specifically the adsorbed Hg on soil colloids, iron/manganese oxide-bound Hg, organic matter-bound Hg, and residual Hg (mainly HgS), differentiated by a selective sequential extraction procedure (Wang et al., 2011, 2019; O'Connor et al., 2019). HgS is the main form of Hg in mining areas (e.g., HgS/THg: 64 −94% in the soils of Wanshan mining area), and an important chemical form of Hg in paddy soils (e.g., HgS/THg: 16.74% in Qingzhen paddy soils) (Bao et al., 2011; Yin et al., 2016; Feng et al., 1997). Given its insolubility, HgS has long been considered to be unavailable in plants and microorganisms, and the formation of HgS is usually an important remediation pathway to deposit Hg in farmlands, through which Hg(II) in sediments and paddy soils is immobilized and the production and bioaccumulation of MeHg are inhibited (Benoit et al., 1999; Li et al., 2018, 2017; Winfrey and Rudd, 2010). Our previous studies showed that sulfur such as elemental sulfur, sulfate, and thiosulfate addition reduced Hg accumulation in rice by promoting the conversion of Hg-dissolved organic matter (DOM) compounds to HgS (Li et al., 2018, 2017). However, HgS particles in contaminated soils have the potential to be available to plants to some extent (Han et al., 2008; Liu et al., 2019). Recent studies showed that HgS can be methylated by anaerobic microorganisms (e.g., sulfate-reducing bacteria) under anaerobic conditions (Benoit et al., 2010; Tian et al., 2021). DOM and sulfate or elemental sulfur were also documented to enhance microbial Hg methylation under sulfidic conditions by preventing HgS precipitation, thereby increasing MeHg bioaccumulation (Li et al., 2019b, 2019a; Ravichandran et al., 1998). However, it is still unknown whether and how bulk HgS can be methylated in paddy soils and accumulated in rice, and what are the key factors that can affect the metabolism of bulk HgS in rice paddies. These issues are still far from understood and deserve further research.

In the present study, the potential methylation capacity and the accumulation of MeHg and THg in rice plants exposed to bulk α-HgS were investigated. The dynamic changes in the production and absorption of MeHg in paddy soils and rice tissues during rice cultivation were examined, and the accumulation of THg and MeHg in rice after two seasons was further discussed. The methylation capacity of bulk α-HgS in mining areas and the newly deposited paddy soils were also compared. This study is expected to provide a comprehensive understanding of the risks of mineral and precipitated Hg forms in croplands.

2. Materials and methods

2.1. Soil collection and rice cultivation

Hainan Province is an important rice growing region in China, where the paddy soils were collected from and investigated as the contrast in this study. The surface soil (0–20 cm) of paddy fields in Hainan Province was collected for rice cultivation and was set as the control. The soil was air-dried and mixed evenly and then was ground and sieved through a 2-mm aperture to remove stones and plant residues. Soil pH and the contents of iron, manganese, Hg, sulfur, and organic matter were characterized. The results are shown in Table S1 (see the Supporting Information, Table SI). Each pot contained 5 kg of the sieved paddy soils. Urea (0.4 g/kg) and K_2HPO_4 (0.4 g/kg) were added to the soils to provide basic nutrition for rice growth. The Hg contaminated soils collected from the Hg mining area (Wanshan District, Guizhou, China) and the newly deposited Hg-contaminated soils (sourced from Hainan Province) were prepared according to previous report (Li et al., 2017). Briefly, Hg-containing solution (Hg^{2+}) was mixed with the soils and incubated for 1 month to reach Hg equilibrium. The final Hg content of newly contaminated soils was 80 mg/kg (Hg/soil). The physicochemical properties of these soils are shown in Table S1. Moreover, to compare with the Hg mining-contaminated soils (the THg and HgS detected ranged from 24.3 to 347.5 mg/kg (Ding et al., 2004) and 15.6–326.7 mg/kg, respectively (HgS/THg: 64 −94% in Hg mining area soils) (Bao et al., 2011)), pot experiments were set to be incubated with 100 mg/kg bulk α-HgS (obtained from commercial cinnabar reagent (Aladdin Co., Shanghai, China), purity: > 99%, ground and passed through a 100-mesh sieve), mimicking the natural environment of Hg mining areas with a moderate HgS level in this study. The soils were flooded and placed to reach HgS equilibrium. After the soils were fully mixed and aged for 1 month, 1 kg of soil was added to special rhizosphere bags (500-mesh root bag, size: 12 cm in diameter and 18 cm in height) to distinguish between the rhizosphere and non- rhizosphere. Another 4 kg of soil was placed outside the rhizosphere bag, as illustrated in Fig. S1. Uniform seedlings (Nongken 7, hybrid japonica) after three weeks of hydroponic cultivation in greenhouse were transplanted into paddy soils, with three duplicates in each group. The pot experiments were conducted outside the building with natural air flow (Fig. S2). During the whole rice growth season, flooded conditions were strictly conducted. The flooded conditions (3 cm of distilled water) were maintained throughout the rice growth season. Each group was evaluated in triplicate. After the first growing season (approximately 4 months for one rice growth cycle), the soils were fully dried and exposed to the natural environment. In the second season (4 more months for another rice growth cycle), the soils from each treatment after the first growing season were dried, smashed, and sieved. Four kilograms of soil was weighted into each plastic bucket. After three weeks of hydroponic cultivation, uniform seedlings were transplanted into the paddy soils. The culture conditions were the same as those in the first experimental season.

Table 1
Treatments in the batch experiments.

Treatment groups	Specific operation
HgS + DI water	6 mg of bulk α-HgS + 20 mL of DI water
HgS + DOM	6 mg of bulkα-HgS + 20 mL of DOM extracted from soil
HgS + DOM+thiosulfate sodium	6 mg of bulk α-HgS + 20 mL of DOM extracted from soil + 20 mM thiosulfate sodium
HgS + thiosulfate sodium + soil	6 mg of bulk α-HgS + 20 mM thiosulfate + 20 mL of DI water sodium + 5 g of Hainan soil (without Hg) + 20 mL of DI water

2.2. Rice harvest

In the seedling, tillering, flowering, and mature stages of rice, the soil pH value of each group was determined during the first season. The rhizospheric and nonrhizospheric soils were fully mixed and collected separately. The rice plants were also sampled in each rice growth period. The tissues of rice plants were divided into roots, straw (leaf and stem), and grain (mature stage). After drying at 40 °C, the biomass of rice tissues was calculated, and then the samples were ground and sieved through a 100-mesh sieve (rice roots without removing iron plaque). The rice tissues of the second season were also collected, dried, ground, and sieved through 100 mesh sieve at the mature stage. The soil samples were placed into 50-mL centrifuge tubes, lyophilized, ground into fine powders, and sieved through a 100-mesh sieve. All samples were stored at − 20 °C before further analysis.

2.3. Batch experiments

Batch experiments were conducted to explore the factors (DOM and reduced sulfur) affecting the dissolution of bulk α-HgS in paddy soils. There were four treatment groups, and each group was designed in triplicate. The disproportionation of thiosulfate can occur under anaerobic and acidic conditions to form reduced S (e.g., S and S^{2-}/HS^-) (Li et al., 2018). Thiosulfate addition was employed here to simulate

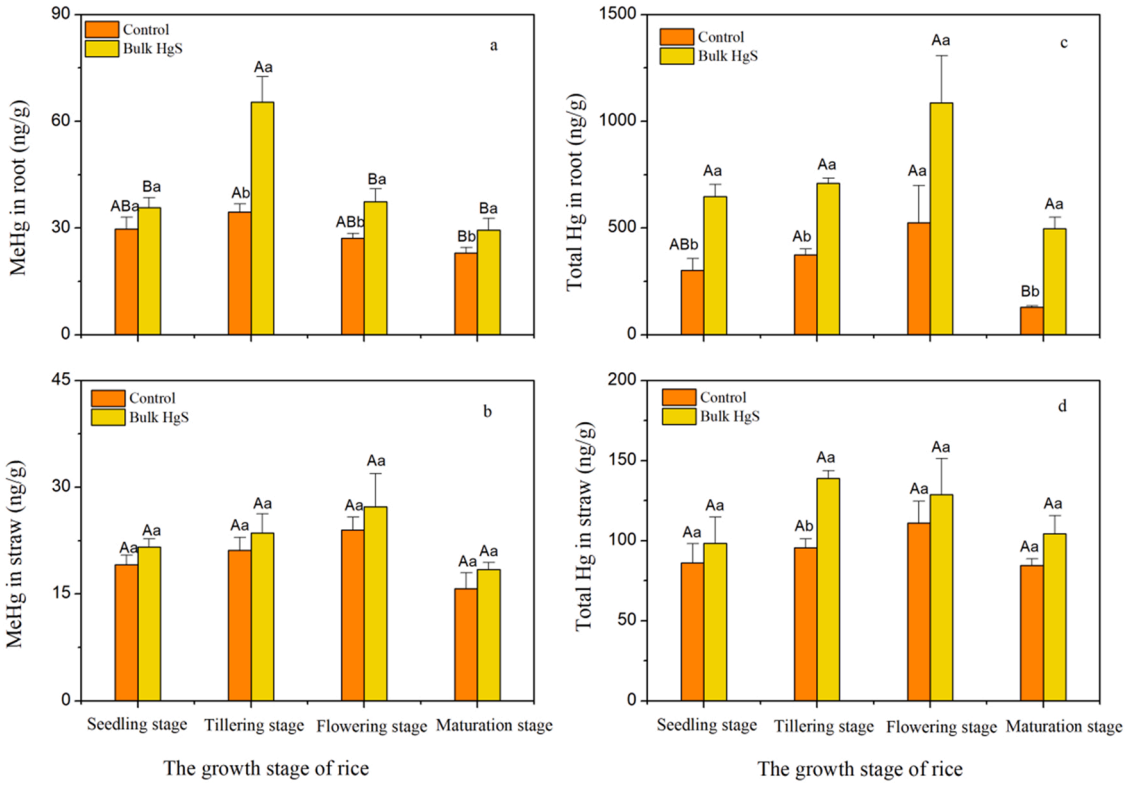


Fig. 1. MeHg concentrations in the roots (a) and straw (b) at different stages. Total Hg concentrations in the roots (c) and straw (d) at different stages. Different lowercase letters above the columns indicate significant differences between the bulk HgS and control groups at the same growth stage ($p < 0.05$). Different capital letters above the columns indicate significant differences between the different growth stages under the same Hg treatment ($p < 0.05$). Error bars are the standard deviations of the means (n = 3).

reduced sulfur conditions in paddy soils because of the instability of reduced sulfur, such as Na_2S and other sulfur compounds. There were four treatments in the batch experiment. Six milligrams of bulk α-HgS was weighted into the centrifuge tube. The treatment of the mixture of bulk α-HgS with 20 mL of DI water was set as the control group. The other three treatments were set as follows: HgS + DOM, HgS + DOM + thiosulfate sodium, and HgS + DOM + soil extracts. The treatments of these four groups are detailed in Table 1. Specifically, 200 g of paddy soils collected from Hainan sieved through a 2-mm sieve was weighed, mixed with 500 mL of DI water, and vibrated for 30 min. After centrifugation for 30 min at 6000 r/min, the supernatant was further passed through a 0.45-μm filter membrane (JinTeng Nylon 66, China) to obtain the soil DOM. All the treatments were centrifuged for 30 min at 6000 r/min and then the supernatant was collected for subsequent dissolved THg determination.

2.4. MeHg and THg analysis

2.4.1. THg analysis

Each sample (50 mg of root or straw; 100 mg of polished gain) was weighed and placed into a Teflon vessel, mixed with 4 mL of HNO_3 (GR) and 0.5 mL of H_2O_2 (GR), sealed and predigested at room temperature overnight. The soil samples (~100 mg) were mixed with 4 mL of HNO_3 (GR), 3 mL of $HClO_4$ (GR), and 3 mL of HF (GR) overnight, and then the Teflon vessels were equipped with stainless-steel containers and digested at 160 °C for 6 h. After digestion, each mixture was concentrated to 0.5−1 mL at 90 °C and diluted to 4 mL with 2% (V/V) HNO_3 and 0.1% (V/V) β-mercaptoethanol. THg was determined by inductively coupled plasma mass spectrometry (ICP-MS, Thermo Elemental X7, USA) in normal mode. Instrument optimization was performed using a standard tuning solution (Bi, 20 ppb) every time. THg in the certified reference materials (CRM) of rice grain (NIES 10) and soil (IAEA-405) was recovered at 96–108% and 91–110% (n = 6), respectively. The calibrating standards of THg were prepared by diluting the Hg standard stock solution (GBW 08617, National Research Centre for CRMs, China) using 2% (V/V) HNO_3 and 0.1% (V/V) β-mercaptoethanol. The method detection limit (MDL) is defined as three times of the standard deviation for a sample (n = 3), which was calculated to approximately 0.1 μg/Kg in this study.

2.4.2. MeHg analysis

The rice and soil samples (~50 mg) were weighed into 15 mL centrifuge tubes (Corning, USA) and mixed with 2 mL 25% (w/w) KOH-CH_3OH. The mixture was shaken at 60 °C for 4 h. After cooling to room temperature, the mixture was diluted to 10 mL with ultrapure water and centrifuged at 4000 rpm/min for 30 min to obtain the supernatant. MeHg in the supernatant was ethylated by $NaBEt_4$ with citrate buffer (pH 4.1–4.5) and analyzed by cold vapor atomic fluorescence spectrometry (GC-CVAFS, Brooks Rand, USA) according to USEPA Method 1630 (USEPA, 2001). MeHg in the reference standard material (estuarine sediment ERM-CC580) was recovered at 92 ± 5% (n = 6). The instrument detection limit (IDL) was defined as three times of the standard deviation of the signal for a reagent blank (n = 12), and three times of the standard deviation for a sample (n = 3) was defined as the method detection limit. The IDL and MDL in this study was calculated to about 0.009 pg and 4 ng/Kg, respectively.

2.5. Soil Hg speciation analysis with XANES

Soil samples at harvest in the first season were ground to fine powders and smeared onto Scotch tape. Hg L_3-edge X-ray absorption near-edge structure spectroscopy (XANES) experiment was carried out on 1W1B beamline at the Beijing Synchrotron Radiation Facility (BSRF). The spectra of soil samples were collected in fluorescence mode between

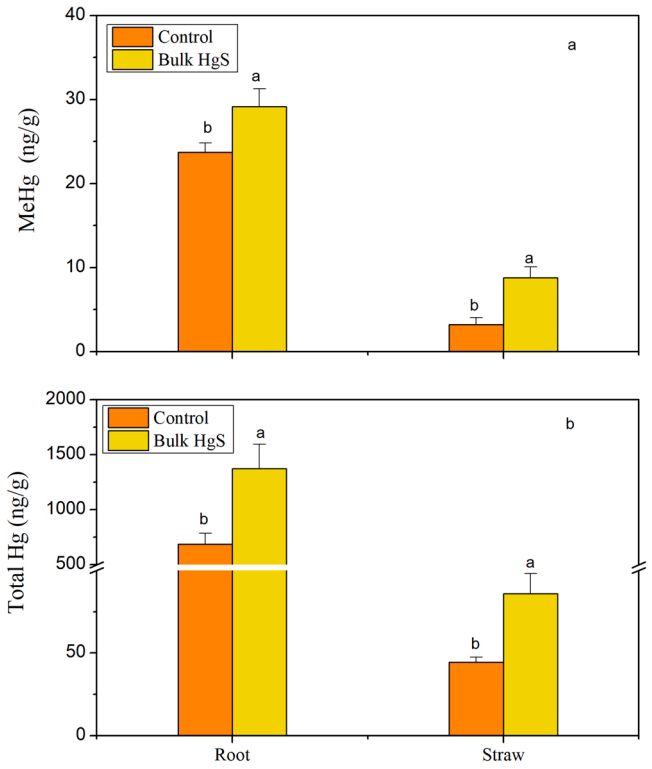


Fig. 2. MeHg (a) and total Hg (b) concentrations in rice roots and straws at the maturation stage collected from the second culture season. Different lowercase letters above the columns indicate significant differences between the bulk HgS treatment and control groups ($p < 0.05$). Error bars are the standard deviations of the means (n = 3).

− 50 and 100 EV relative to Hg L_3-edge energy, while those of the reference materials (α-HgS, $HgCl_2$, Hg-glutathione ($Hg(GS)_2$), CH_3HgCl, CH_3HgGS) were collected in transmission mode. All data were normalized and analyzed using ATHENA software. Linear combination fitting (LCF) was performed to determine the relative percentage of each potential species of Hg in these samples.

2.6. Data analysis

Variance analysis and Student's *t*-tests were conducted using SPSS 20.0 software (SPSS Inc., Chicago, IL, USA) to identify significant differences ($p < 0.05$) among treatments. All data are shown as the average ± standard deviation (SD) in all graphs and tables.

3. Results

3.1. MeHg and THg accumulation in rice

The biomass of rice tissues at harvest, which indicated the non-phytotoxicity of bulk α-HgS on rice, is shown in Fig. S3 (first season) and Fig. S4 (second season). In the first growing season, the biomass of rice roots, straw, and grain was not markedly impacted by bulk α-HgS exposure (Fig. S3) ($p > 0.05$). The biomass of rice grain harvested after the second growth season with bulk α-HgS treatment declined slightly when compared to the control (Fig. S4).

The concentrations of MeHg in rice roots and straw at different growth stages in the first growing season are shown in Figs. 1a and 1b. Bulk α-HgS spiking significantly increased the MeHg concentrations in the roots when compared with that of the control group, especially for roots of tillering stage with a 50% higher rate than that of other growth stages. The trend of average root MeHg at different growth stages under bulk α-HgS exposure followed the order of tillering stage > flowering stage > seedling stage > maturation stage (Fig. 1a). There was no significant difference in root MeHg among the different growth stages of the control group. MeHg in rice straw of bulk α-HgS exposed group was a bit more than that of the control, and approximately half of that in roots (Fig. 1b). Notably, the MeHg concentrations in roots and straw of the maturation stage declined in comparison to those of the tillering and flowering stages. In contrast to MeHg, the THg accumulation in rice roots and straw was much higher, especially in roots (10- to 20- fold higher than the MeHg accumulation). Moreover, the THg accumulation in roots of the maturation stage with bulk α-HgS addition was approximately 5 fold greater than that of the control groups (Fig. 1c). When comparing the transportation ratios of MeHg (the percentage of straw MeHg content/root MeHg content) with THg (the percentage of straw THg content/root THg content) (36%−73% for MeHg vs. 12−21% for THg), we can conclude that MeHg is much more prone to be transported from root to aboveground parts than that of inorganic Hg. However, bulk α-HgS exposure enhanced the accumulation of Hg in rice roots, especially for THg in the maturation stage, suggesting a high risk of Hg release in the rhizosphere in buk α-HgS containing croplands.

The second growth season of rice was planted to explore the continued risk of bulk α-HgS. MeHg concentrations in rice roots and straw at the maturation stage in the second growing season are shown in Fig. 2a. In the control, the concentration of MeHg in roots and straws was 23.70 ± 1.12 μg/kg and 3.19 ± 0.84 μg/kg, respectively. The exposure of bulk α-HgS markedly increased the concentrations of MeHg in roots and straws, and the increasing rates were 176.18% (straw MeHg) and 23.04% (root MeHg). Bulk α-HgS addition also significantly enhanced the accumulation of THg in rice plants during the second growing season. The concentrations of THg in rice roots and straws were 1.55 and 1.94 times that of the control group, respectively (Fig. 2b).

Most importantly, bulk α-HgS exposure markedly enhanced the concentrations of MeHg and THg in rice grains in the first and second growing seasons (Fig. 3a and Fig. 3b). Although the MeHg concentration in grains of bulk α-HgS-exposed rice was much lower than that of rice growing in soil from the Hg mining area or newly deposited Hg-contaminated soil, it was approximately 1.5 (first season) and 5 (second season) times that of the control group (Fig. 3a). The grain MeHg concentration in the second growing season was much higher than that in the first season. Similar to MeHg, bulk α-HgS exposure also increased the THg content in rice grains (3−4-fold higher than that of the control) (Fig. 3b). The ratio of MeHg to THg in rice grains of the first growing season with bulk α-HgS addition (43.2%) was slightly lower than that observed in rice grown in the mining area soils; however, the ratio was much higher in the second growing season (67.2%), indicating high MeHg risks in grains after two seasons of rice cultivation with bulk α-HgS exposure (Fig. 3c). Moreover, Hg species and the aging time of newly deposited Hg in contaminated paddy soils are responsible for the accumulation of MeHg and THg in rice grains. Bulk α-HgS has long been considered to be not bioavailable and is believed to be an ideal remediation approach in Hg-contaminated soils. However, the relatively high concentration of DOM, reduced sulfur and high activity of microbes in paddy soils may facilitate the dissolution and bioavailability of bulk α-HgS. The watering conditions (resulting in anaerobic environments) could be another factor influencing the bioavailability of bulk α-HgS and impact the aging time of deposited Hg in rice paddies. Given the abovementioned findings, bulk α-HgS should be considered when assessing Hg risk and developing an efficient remediation approach in Hg-contaminated paddy fields.

3.2. Hg speciation and MeHg content in paddy soils

Net MeHg production in rhizospheric and nonrhizospheric soils at different rice growth stages was determined to reflect the methylation of dissolved bulk α-HgS in paddy fields. In the first growth season, the MeHg content in rhizosphere soils of the bulk α-HgS exposed group was significantly increased compared with that of the control group and

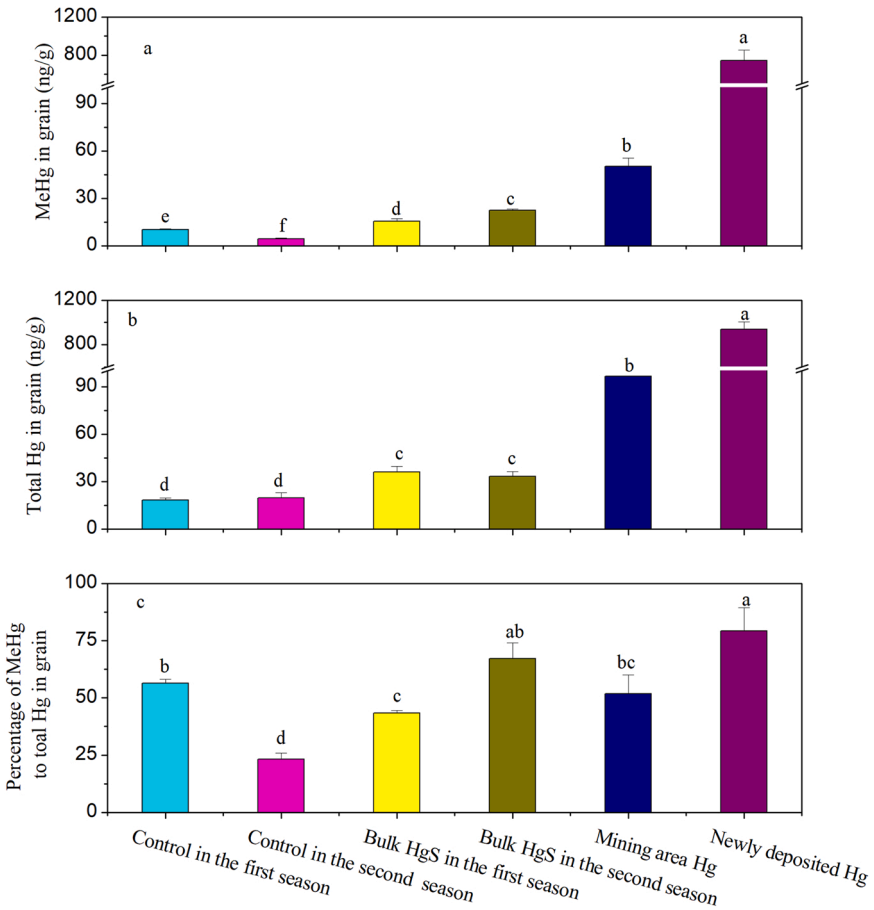


Fig. 3. MeHg (a) and total Hg (b) concentrations in grain under different treatments. The percentage of MeHg to total Hg in grain (c). Different capital letters above the columns indicate significant differences ($p < 0.05$). Error bars are the standard deviations of the means (n = 3).

exhibited an increasing trend following the four rice growth stages. Specifically, the MeHg content in the rhizosphere of the bulk α-HgS treatment was 80.15% and 108.7% higher than those of the control group in the seedling and maturation stages, respectively (Fig. 4a). A similar phenomenon was also observed in the nonrhizospheric soils. The rates of increase for MeHg in the nonrhizospheric soil of the bulk α-HgS treatment compared with that of the control were 35.2%− 134.7% from the seedling stage to the maturation stage (Fig. 4b). The MeHg contents in nonrhizospheric soils at different stages were slightly lower than those observed in rhizospheric soils for both the control and bulk α-HgS treatments ($p > 0.05$), suggesting the elevated methylation of bulk α-HgS with rice cultivation. The concentration of MeHg in paddy soils of the second rice growth season was significantly higher than that of the control, for instance, the MeHg content in rgizospheric soils with bulk α-HgS addition was approximately 2-fold greater than that of the control at the maturation stage of rice growth (Fig. 4c).

The Hg L_3-edge XANES spectra of the rhizosphere soil and standard reference compounds, as well as the linear combination fitting (LCF) data, can directly reflect the chemical forms of Hg in rhizospheric soil from different treatments (shown in Fig. 5). Hg species in the RS-Hg-SR form (represented by $Hg(GS)_2$) accounted for approximately 5.8%, 25.6%, and 77.0% of the total Hg in rhizospheric soils of the bulk α-HgS spiked, mining area Hg, and newly deposited Hg treatments, respectively. The HgS form in rhizospheric soils accounted for 94.2%, 73.4%, and 12.4% of the total Hg in the bulk α-HgS, mining area Hg, and newly deposited Hg treatments, respectively. Hg-S^- structure is generally produced by the combination of dissolved Hg and HS-containing compounds and is usually presented in sulfur-enriched circumstances, such as in rice paddies and rice tissues (Li et al., 2017; Skyllberg and Drott, 2010; Skyllberg, 2015). RS-Hg-like species are prone to be assimilated and methylated by microbes and then available to plants (Graham et al., 2017). HgS nanoparticles are currently reported to be methylated by microbes in rice paddies (Liu et al., 2019), and the methylation potential is dependent on their crystal structures and is enhanced with the existence of natural ligands (Tian et al., 2021). In the present work, significant methylation of bulk α-HgS in the rhizosphere was observed in paddy soil, which resulted in enhanced accumulations of MeHg and THg in rice, especially in grains, at the maturation stage. Taken together, we suggest adding bulk α-HgS, along with its nano-/micro-scaled counterparts, to the risk assessment practice in Hg-contaminated rice paddies and DOM-enriched circumstances, although HgS has long been expected to be stable and not bioavailable.

3.3. *Dissolution of bulk α-HgS*

Batch experiments were designed to explore the factors influencing the dissolution of bulk α-HgS in paddy soil. The concentration of dissolved Hg from bulk α-HgS was 5.6 ± 1.2 μg/L in deionized (DI) water for 24 h (control group) (Fig. 6). The addition of DOM significantly enhanced the dissolution of bulk α-HgS when compared with the control group (28.60 μg/L). Supplementation with thiosulfate further improved the dissolution of bulk α-HgS compared with that of the bulk-HgS+DOM group. A similar phenomenon with a relatively high dissolution rate of bulk α-HgS (51.3 μg/L) was also observed in the soil extract experiment with thiosulfate addition, which further indicated the positive effects of DOM and thiosulfate on Hg release from bulk α-HgS.

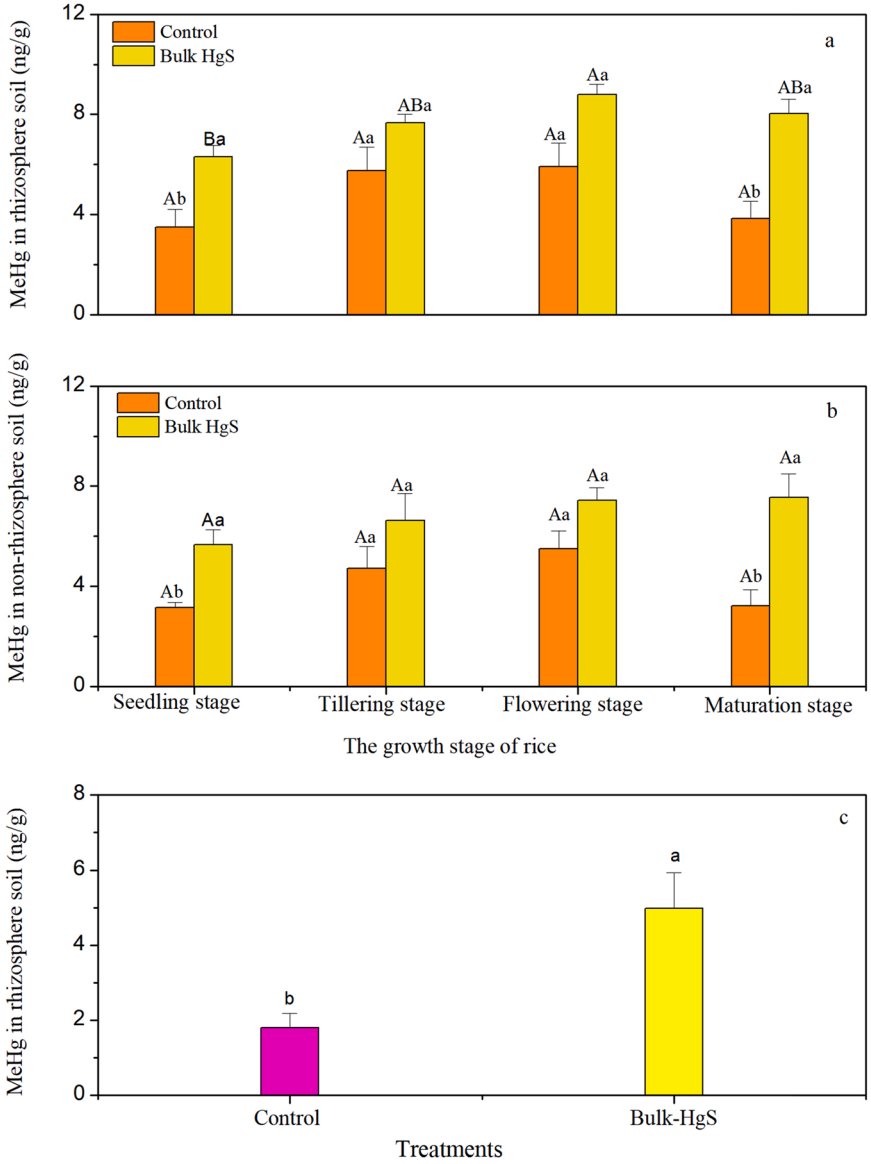


Fig. 4. MeHg contents in the rhizosphere (a) and non-rhizosphere (b) soil at different stages of the first season. MeHg contents in rhizosphere soil in the second season (c). Different lowercase letters above the columns indicate significant differences between the HgS and control groups at the same growth stage ($p < 0.05$). Different capital letters above the columns indicate significant differences between the different growth stages under the same Hg treatment ($p < 0.05$). Error bars are the standard deviations of the means (n = 3).

4. Discussion

4.1. Hg accumulation in rice

The concentrations of THg in rice plants under bulk α-HgS exposure were significantly higher than those of the control group. The THg levels in rice grains (36.0 μg/kg in the first season, 36.3 μg/kg in the second season) were much higher than the Chinese national limit for cereals (20 μg/kg, GB2762–2017), which implies that bulk α-HgS in paddy fields still gives rise to environmental and health risks. Similar results were also reported by other researchers in sediments. MeHg contents in HgS particles containing soil (5 and 50 mg/kg Hg) were 2−4 times higher than thoset in the control, which suggested potential assimilation of HgS particles by plants (Liu et al., 2019). Another study showed that HgS particles in soil could be more prone to dissolve after three seasons of planting (Han et al., 2008). The present work showed that 5.8% of bulk α-HgS in rhizospheric soil was transformed into RS-Hg-SR (Hg $(GS)_2$)-like forms with rice cultivation after the first growing season.

Similar to THg, the concentration of MeHg in rice tissues of all growth stages under bulk α-HgS exposure (mimicking the contaminated areas, such as in Wanshan District) was much higher than that of the control. In waterlogged paddy soils, the relatively low pH and redox potential induced by the anaerobic condition and high DOM value could lead to dissolution of mercuric sulfide, increase the activity of Hg-methylation bacteria, and then elevate the concentrations of MeHg in soil and rice (Zheng et al., 2019; Drott et al., 2007; Wang et al., 2021a, 2021b). Our study quantified the contribution of bulk α-HgS in paddy soils to the accumulation of MeHg and THg in rice. Fortunately, the current results revealed that the proportion of MeHg to THg in rice tissues declined in the second rice culture season compared with the first rice culture season (0.2% for roots and 15.8% for straw in the second rice culture season vs. 5.9% for root and 17.7% for straw in the first culture season). The ratio of MeHg to THg in rice grains was also observed to decrease with the increasing bulk α-HgS in paddy soils (Fig. 3c), which indicated that the conversion of dissolved Hg to bulk α-HgS in paddy fields could reduce the MeHg/THg value, thus alleviating the health risks of MeHg to some extent. However, it is notable that paddy fields are of specific environments with complex redox conditions and physicochemical properties which facilitate the conversion of diverse Hg species (Wang et al., 2021a, 2021b). The uptake, transportation, transformation, and accumulation of various Hg species in rice paddies based on field studies still need to be further explored.

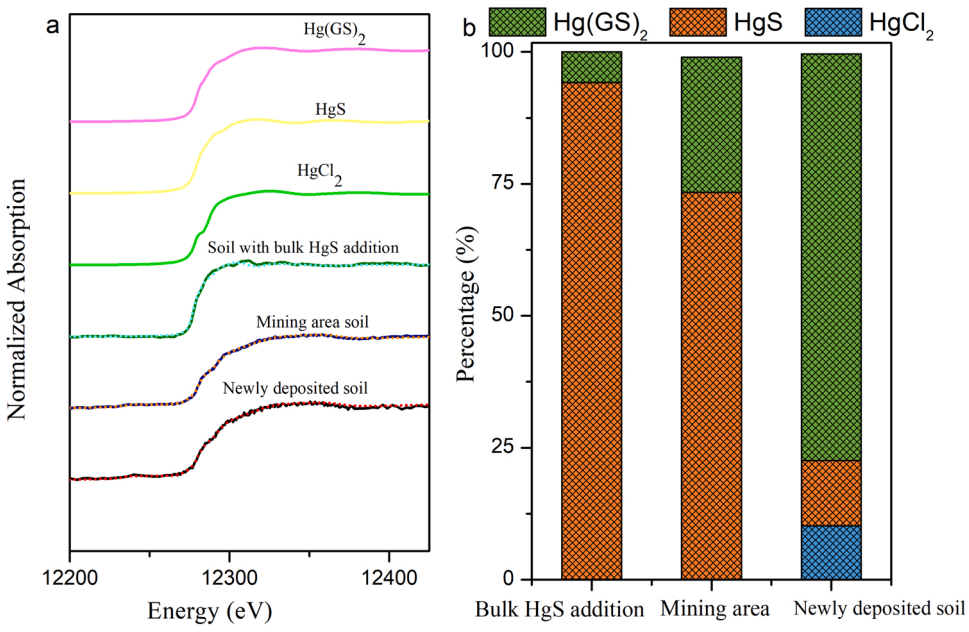


Fig. 5. (a) Normalized Hg L_3-edge SR-XANES spectra of the reference compounds and three soil samples from the soil with bulk HgS addition, soil collected from Wanshan mining areas and newly deposited soil (soil amended with $HgCl_2$), and linear combination fitting (LCF) results for the three soil samples. The dotted lines indicate the fitting data.

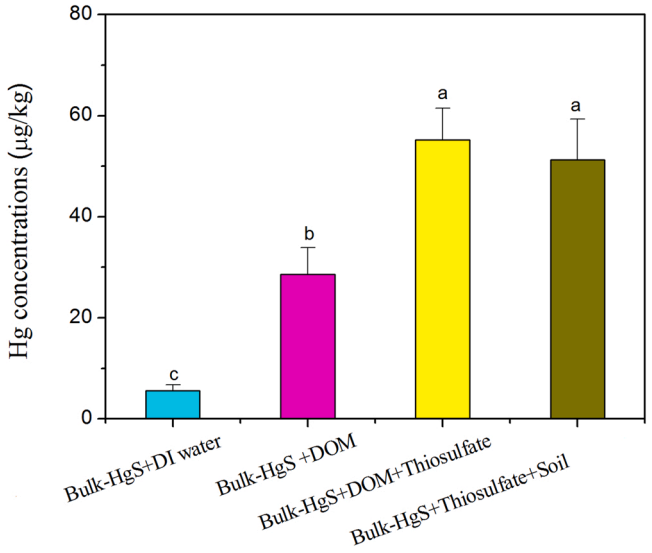


Fig. 6. The dissolution of bulk HgS in different treatments. Different lowercase letters above the columns indicate significant differences between the bulk HgS and control groups at the same growth stage ($p < 0.05$). Error bars are the standard deviations of the means (n = 3).

4.2. Change of MeHg in soil

Although HgS has relatively low solubility ($logK_{sp} \approx -52.1$), the health risks induced by bulk α-HgS in specific and complex natural environments, such as in paddy fields, should attract more attention from the present work due to the potential methylation and dissolution of bulk α-HgS with rice cultivation. It has been previously reported that the order of methylation rate constants for Hg follows β-HgS < α-HgS < FeS-Hg^{II} < NOM-Hg^{II} <$Hg(NO_3)_2$ (aq) (Jonsson et al., 2012). The dissolution of HgS could be affected by DOM or reduced sulfur (Li et al., 2019b; Paquette and Helz, 1997; Ravichandran et al., 1998; Waples et al., 2005). HgS particles can be converted to soluble and bioavailable Hg-complexes (HgS_2^{2-}, $HgHS_2^{2-}$, $Hg(HS)_2$, etc.) in the presence of reduced sulfur (Lei et al., 2021; Li et al., 2019b, 2019a). Moreover, the dissolution rate of HgS mediated by DOM was correlated with the aromaticity and molecular weight of the organic matter (Waples et al., 2005). Therefore, DOM and reduced sulfide in paddy fields play critical roles in mobilizing bulk α-HgS and promoting the methylation of dissolved Hg which would result in elevated THg/MeHg accumulation in rice.

The oxidized rhizosphere during alternating watering and drainage conditions in rice paddies provides living sites for a rich diversity of microorganisms, as well as high sulfate reduction and sulfide reoxidation. In anoxic environments, sulfate-reducing microorganisms (SRMs) are considered the primary methylators of inorganic Hg (Kerin et al., 2007; Schaefer and Morel, 2009). The bioavailable carbon is an important driver of MeHg production in wetlands (Windham-Myers et al., 2014), and is also suitable for paddy fields. In waterlogged rice fields, root-derived organic matter, especially acetate (an index of labile C pools), contributed to the elevated soil MeHg concentrations (Zhao et al., 2018). Moreover, a previous study showed that soil MeHg had a positive correlation with the activities of SRM and the contents of organic matter, sulfate, and THg (Rothenberg and Feng, 2012; Zhao et al., 2018). In the present study, the MeHg contents in rhizospheric soil at all rice growth stages were much higher than those in the nonrhizosphere soil, which might be partly attributed to the activated methylation microbes (e.g., SRM) and the enhanced release of dissolved Hg from bulk α-HgS induced by root activity and root-derived organic matter in the rhizospheric soil.

Sulfate, SRM, organic matter, iron, manganese, and pH should play key roles in modulating the formation of MeHg in paddy fields (Lei et al., 2021; Rothenberg and Feng, 2012; Xing et al., 2020). Specifically, stronger reducing conditions promoted the dissolution of Fe/Mn oxides, which correspondingly increased Fe and Mn concentrations in soil pore water (Guo et al., 2018). Fe concentrations are tightly correlated with HgS dissolution and are considered one of the most effective factors in controlling the stability of HgS (Wang et al., 2013). It was also reported that the contents of soluble carbon, iron, and manganese in the pore water of rice paddies in the early rice growth stage were higher than those in the later growth stage, and rhizosphere Eh first increased,

decreased, and then increased at the heading stage (Guo et al., 2018). Therefore, the stability of MeHg in paddy fields with rice cultivation is highly affected by dynamic changes in pH, Eh, Fe, and soluble carbon (Frohne et al., 2012; Beckers et al., 2019). In this study, whether in the rhizosphere or nonrhizosphere soil, the concentration of MeHg in paddy soils followed the order of flowering stage > maturation stage > tillering stage > seedling stage, which provides clues by taking efficient farming measures, such as crop rotation, irrigation and selection of fertilizers at different rice growth periods, to control the production of MeHg in HgS-containing croplands and THg/MeHg accumulation in rice grains.

5. Conclusions

Because of its relative stability, bulk α-HgS has long been considered not bioavailable although its nanoscale counterparts were reported to be methylated by microbes. Rice paddies are characterized as a complex environment with alternating watering and drainage regimes, dynamically changed pH, Eh, DOM, anaerobic and aerobic conditions and rice cultivation, providing special areas for Hg conversion. In the present study, bulk α-HgS at the mining contamination level did not impact rice growth but was found to be available and methylated in the rhizosphere with rice cultivation. The dissolved and methylated Hg species can subsequently be transported and accumulate in rice and lead to overloads of THg and MeHg in rice grains, although the proportion of MeHg to THg in grains is lower than that from dissolved Hg (e.g., Hg^{2+} and DOM-Hg) contaminated paddy soils. Sulfur compounds such as DOM, soil extract, and thiosulfate facilitate the dissolution of bulk α-HgS in paddy soils. This study offers new insights regarding the bioavailability of precipitated bulk α-HgS in paddy fields, especially with rice cultivation. Although increasing the proportion of HgS in soils as a remediation method for Hg-contaminated fields is helpful for reducing its activity, the health risks induced by HgS mobilized by rice culture should be considered and still need more assessments. Moreover, these findings showed that a small part of bulk α-HgS was converted into organic matter-bound Hg (RS-Hg-SR ($Hg(GS)_2$)) after one cropping of rice, but whether this conversion ratio will further increase after multiple cropping still requires further study.

CRediT authorship contribution statement

All the contributors of this work have approved the final version of this manuscript, and declare no financial and personal relationships with other people or organizations that can influence this work if this paper could be accepted.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jhazmat.2021.127394.

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